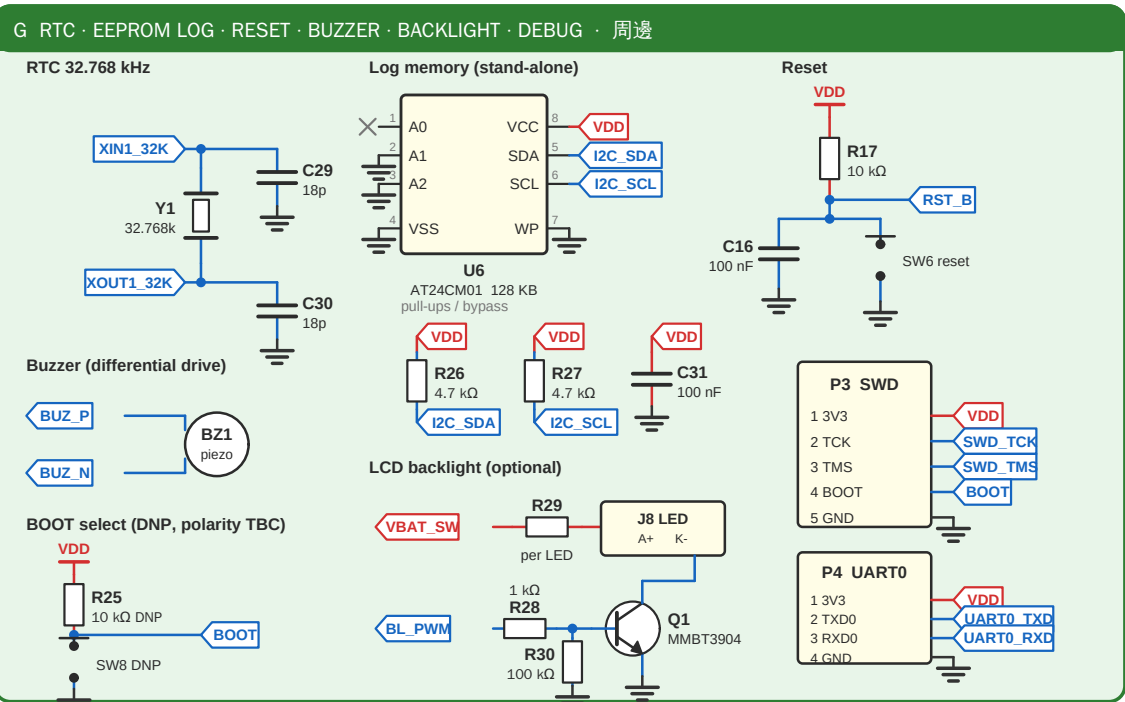
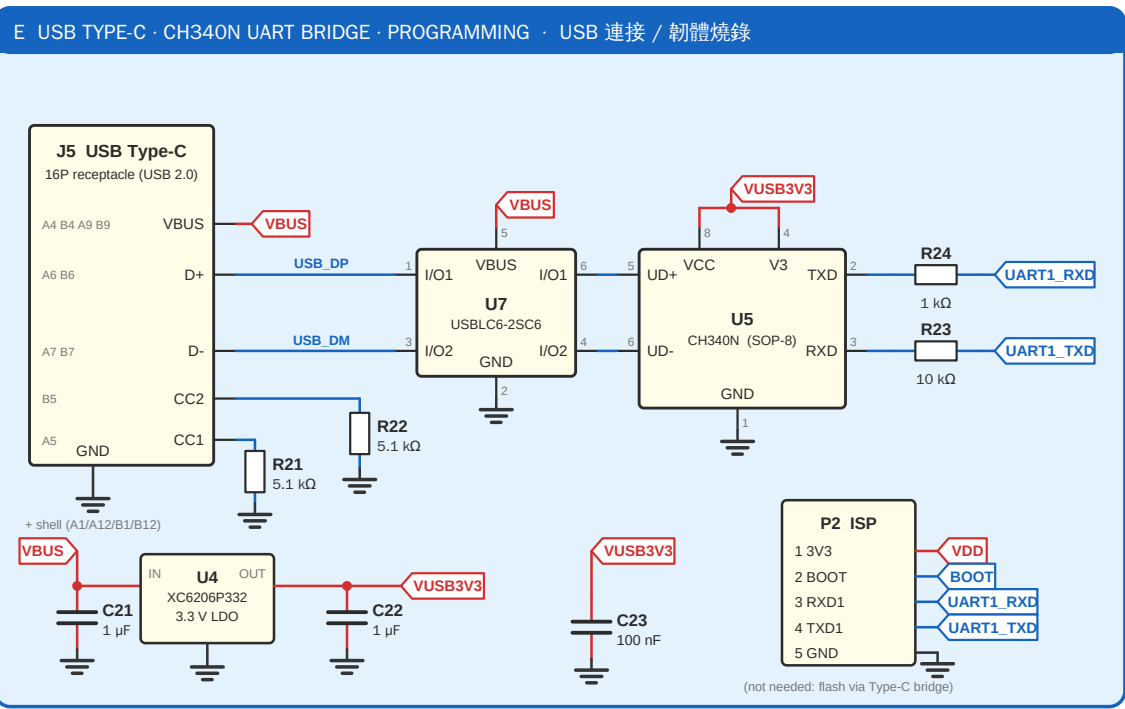
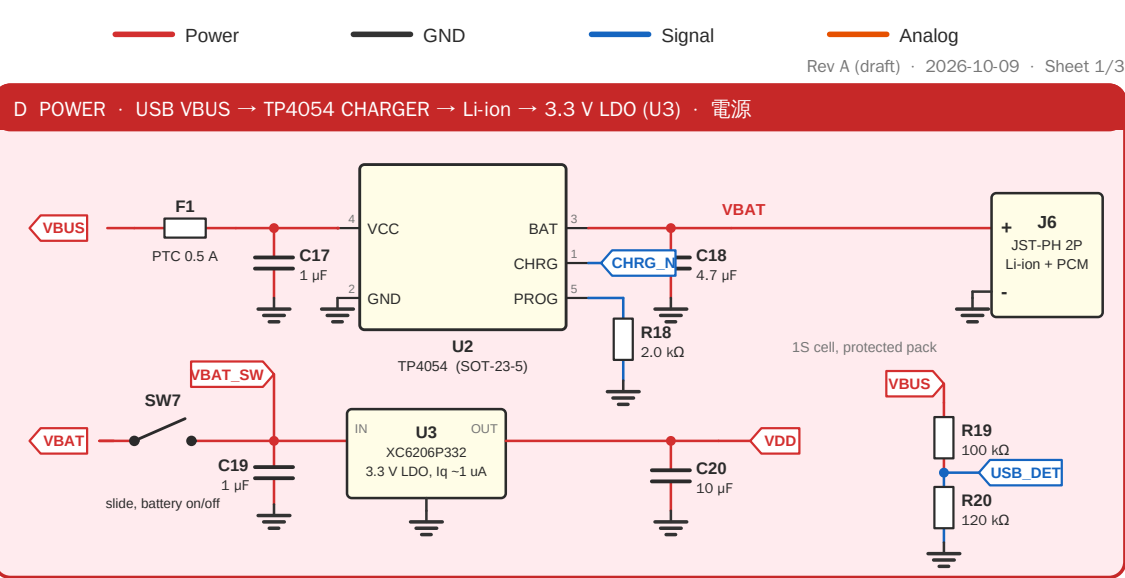
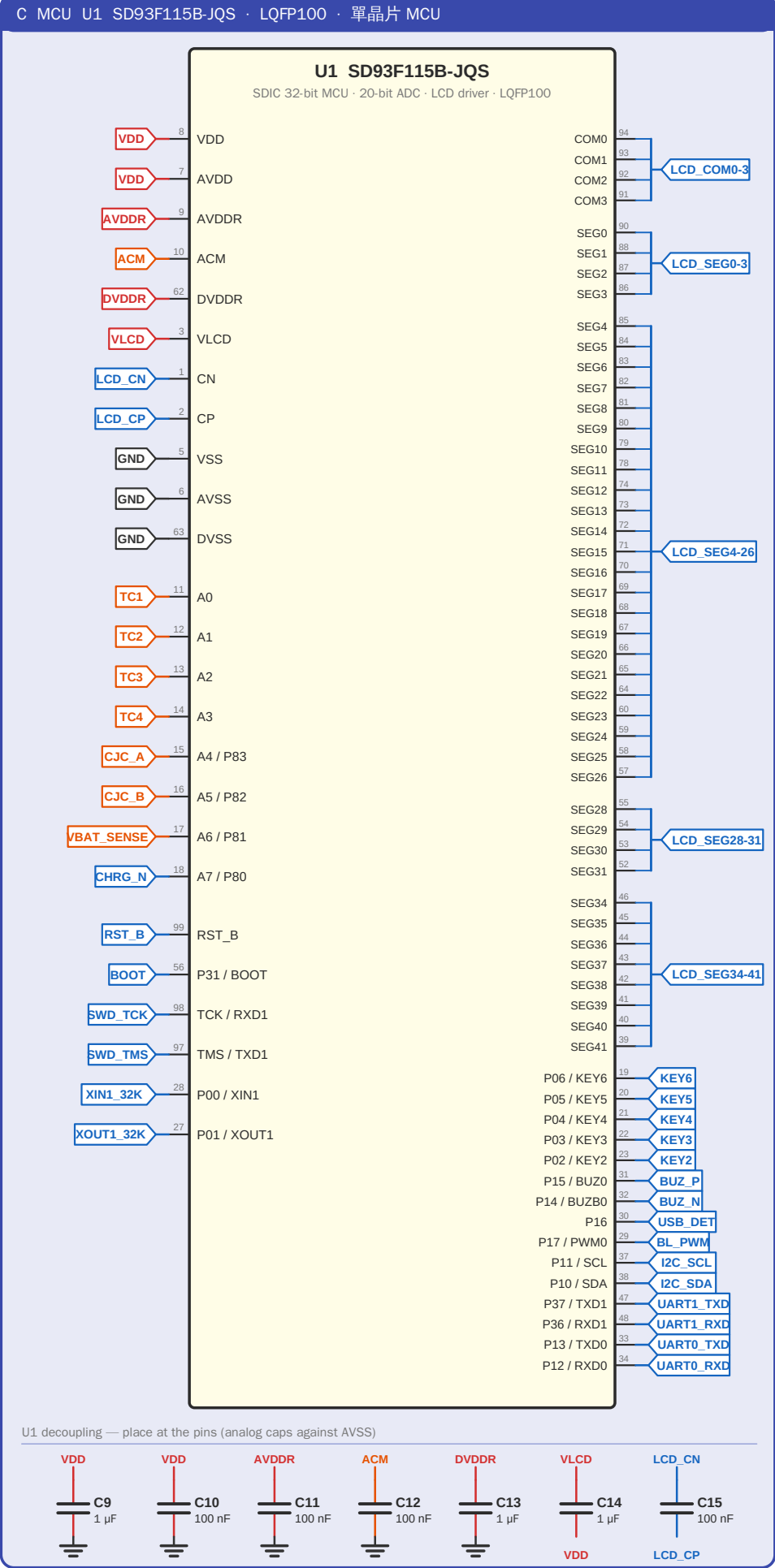
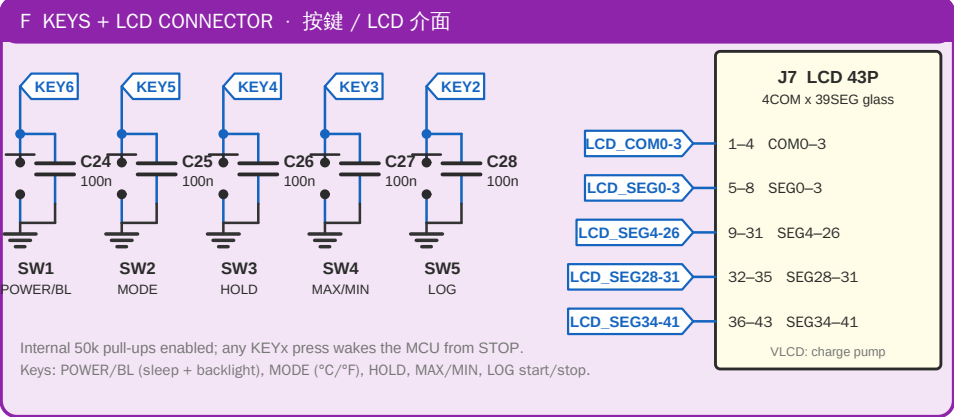
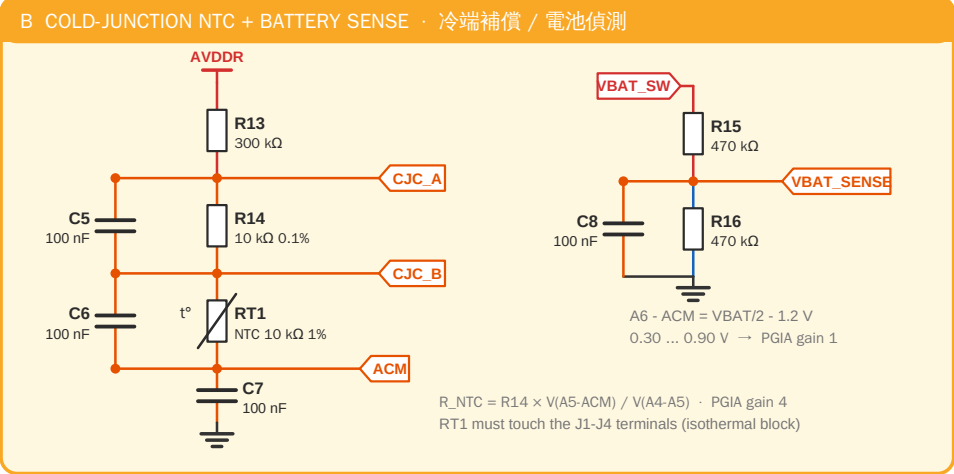
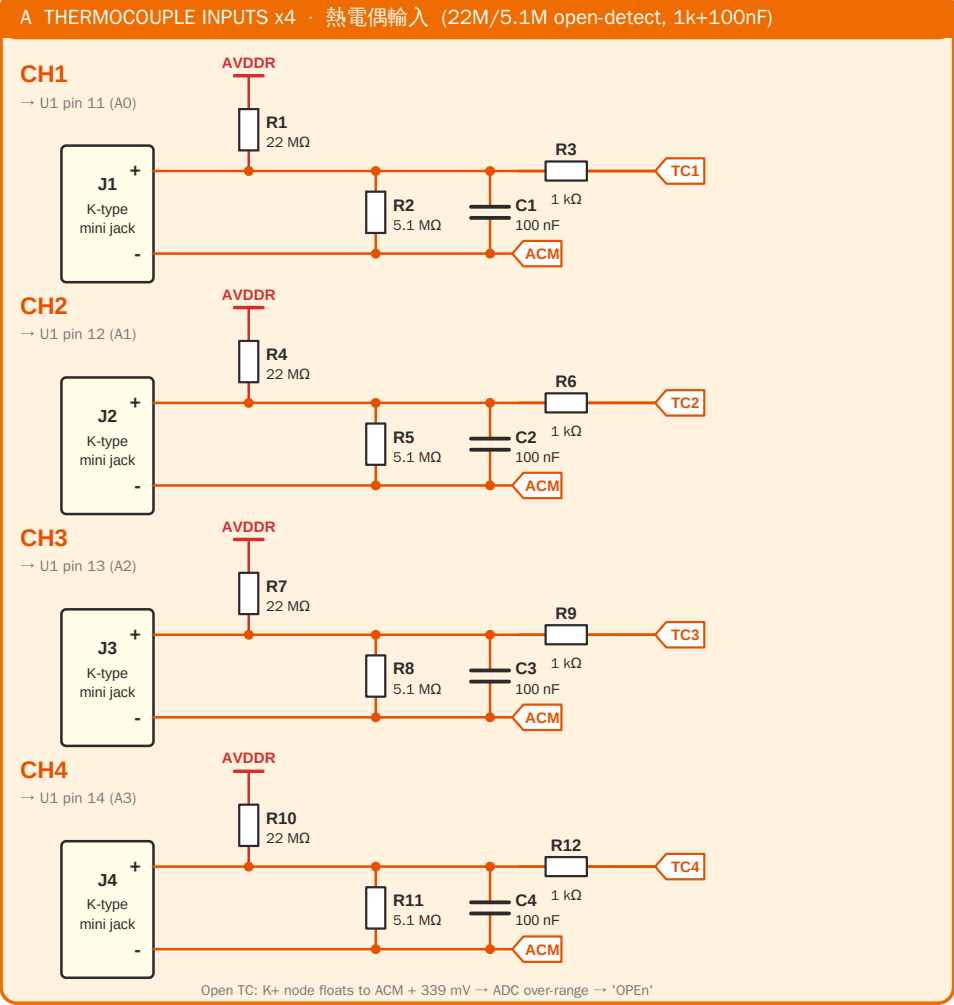


4-CH K-TYPE THERMOMETER / LOGGER · 四通道 K 型溫度計 / 溫度記錄器

MCU SD93F115B-JQS (LQFP100) · 4 x K-type thermocouple, SD-ADC + PGIA · USB Type-C UART logger · 1S Li-ion + charger



U1 SD93F115B-JQS (LQFP100) — PIN ASSIGNMENT · 腳位分配

Datasheet v0.1 pin names → net / function in this design. NC pins must stay floating.

1	CN LCD_CN (C15)	26	NC NC - leave floating	51	NC NC - leave floating	76	NC NC - leave floating
2	CP LCD_CP (C15)	27	P01/KEY1/XOUT1 XOUT1 (Y1)	52	P35/SEG31/CLKout LCD SEG31 -> J7.35	77	NC NC - leave floating
3	VLCD VLCD (C14 to VDD)	28	P00/KEY0/XIN1 XIN1 (Y1)	53	P34/SEG30/CCP LCD SEG30 -> J7.34	78	P51/SEG11 LCD SEG11 -> J7.16
4	NC NC - leave floating	29	P17/PWM0/PDM0 BL_PWM (Q1)	54	P33/SEG29/BUZ1 LCD SEG29 -> J7.33	79	P50/SEG10 LCD SEG10 -> J7.15
5	VSS GND	30	P16/PWM1/PDM1 USB_DET (R19/R20)	55	P32/SEG28/BUZB1 LCD SEG28 -> J7.32	80	P67/SEG9 LCD SEG9 -> J7.14
6	AVSS GND	31	P15/BUZ0 BUZ_P (BZ1)	56	P31/SEG27/BOOT BOOT (P2, P3, SW8)	81	P66/SEG8 LCD SEG8 -> J7.13
7	AVDD VDD	32	P14/BUZB0 BUZ_N (BZ1)	57	P30/SEG26/LBTIN LCD SEG26 -> J7.31	82	P65/SEG7 LCD SEG7 -> J7.12
8	VDD VDD (C9, C10)	33	P13/TXD0 UART0_TXD (P4)	58	P47/SEG25/AUD LCD SEG25 -> J7.30	83	P64/SEG6 LCD SEG6 -> J7.11
9	AVDDR AVDDR (C11)	34	P12/RXD0 UART0_RXD (P4)	59	P46/SEG24/DAO LCD SEG24 -> J7.29	84	P63/SEG5 LCD SEG5 -> J7.10
10	ACM ACM (C12)	35	NC NC - leave floating	60	P45/SEG23 LCD SEG23 -> J7.28	85	P62/SEG4 LCD SEG4 -> J7.9
11	A0 TC1 (J1 via R3)	36	NC NC - leave floating	61	NC NC - leave floating	86	P61/COM7/SEG3 LCD SEG3 -> J7.8
12	A1 TC2 (J2 via R6)	37	P11/SEG43/SCL I2C_SCL (U6, R27)	62	DVDDR DVDDR (C13)	87	P60/COM6/SEG2 LCD SEG2 -> J7.7
13	A2 TC3 (J3 via R9)	38	P10/SEG42/SDA I2C_SDA (U6, R26)	63	DVSS GND	88	P77/COM5/SEG1 LCD SEG1 -> J7.6
14	A3 TC4 (J4 via R12)	39	P27/SEG41/CS LCD SEG41 -> J7.43	64	P44/SEG22 LCD SEG22 -> J7.27	89	NC NC - leave floating
15	A4/P83 CJC_A (R13/R14 node)	40	P26/SEG40/MOSI LCD SEG40 -> J7.42	65	P43/SEG21 LCD SEG21 -> J7.26	90	P76/COM4/SEG0 LCD SEG0 -> J7.5
16	A5/P82 CJC_B (R14/RT1 node)	41	P25/SEG39/MISO LCD SEG39 -> J7.41	66	P42/SEG20 LCD SEG20 -> J7.25	91	P75/COM3 LCD COM3 -> J7.4
17	A6/P81 VBAT_SENSE (R15/R16)	42	P24/SEG38/SCK LCD SEG38 -> J7.40	67	P41/SEG19 LCD SEG19 -> J7.24	92	P74/COM2 LCD COM2 -> J7.3
18	A7/P80 CHRG_N (U2 pin 1)	43	P23/SEG37/T0CK LCD SEG37 -> J7.39	68	P40/SEG18 LCD SEG18 -> J7.23	93	P73/COM1 LCD COM1 -> J7.2
19	P06/KEY6 KEY6 - POWER/BL (SW1)	44	P22/SEG36/PFD LCD SEG36 -> J7.38	69	P57/SEG17 LCD SEG17 -> J7.22	94	P72/COM0 LCD COM0 -> J7.1
20	P05/KEY5 KEY5 - MODE (SW2)	45	P21/SEG35/INT0 LCD SEG35 -> J7.37	70	P56/SEG16 LCD SEG16 -> J7.21	95	P71/XOUT2 (unused)
21	P04/KEY4/ADC0 KEY4 - HOLD (SW3)	46	P20/SEG34/INT1 LCD SEG34 -> J7.36	71	P55/SEG15 LCD SEG15 -> J7.20	96	P70/XIN2 (unused)
22	P03/KEY3/ADC1 KEY3 - MAX/MIN (SW4)	47	P37/SEG33/TXD1 UART1_TXD -> U5 RXD (R23)	72	P54/SEG14 LCD SEG14 -> J7.19	97	TMS/TXD1 SWD_TMS (P3)
23	P02/KEY2/ADC2 KEY2 - LOG (SW5)	48	P36/SEG32/RXD1 UART1_RXD <- U5 TXD (R24)	73	P53/SEG13 LCD SEG13 -> J7.18	98	TCK/RXD1 SWD_TCK (P3)
24	NC NC - leave floating	49	NC NC - leave floating	74	P52/SEG12 LCD SEG12 -> J7.17	99	RST_B RST_B (R17, C16, SW6)
25	NC NC - leave floating	50	NC NC - leave floating	75	NC NC - leave floating	100	NC NC - leave floating

Green = used · Grey = NC (leave floating) · Amber = available GPIO/SEG not used in this design.

SEG27 (BOOT), SEG32/33 (UART1 ↔ USB bridge) and SEG42/43 (I2C EEPROM) are reserved, leaving 39 SEG lines for the LCD.

LCD GLASS PROPOSAL — 4 COM × 39 SEG · LCD 玻璃規劃

1/3 bias, 4-COM multiplex, VLCD from internal charge pump. Share this table with the LCD glass maker.

J7	MCU pin	SEG	Function	COM0	COM1	COM2	COM3
5	90	SEG0	CH1 sign/label	minus	CH1 label	OPEN/ERR	spare
6	88	SEG1	CH1 digit 1 (A)	a	f	e	d
7	87	SEG2	CH1 digit 1 (B)	b	g	c	DP
8	86	SEG3	CH1 digit 2 (A)	a	f	e	d
9	85	SEG4	CH1 digit 2 (B)	b	g	c	DP
10	84	SEG5	CH1 digit 3 (A)	a	f	e	d
11	83	SEG6	CH1 digit 3 (B)	b	g	c	DP
12	82	SEG7	CH1 digit 4 (A)	a	f	e	d
13	81	SEG8	CH1 digit 4 (B)	b	g	c	DP
14	80	SEG9	CH2 sign/label	minus	CH2 label	OPEN/ERR	spare
15	79	SEG10	CH2 digit 1 (A)	a	f	e	d
16	78	SEG11	CH2 digit 1 (B)	b	g	c	DP
17	74	SEG12	CH2 digit 2 (A)	a	f	e	d
18	73	SEG13	CH2 digit 2 (B)	b	g	c	DP
19	72	SEG14	CH2 digit 3 (A)	a	f	e	d
20	71	SEG15	CH2 digit 3 (B)	b	g	c	DP
21	70	SEG16	CH2 digit 4 (A)	a	f	e	d
22	69	SEG17	CH2 digit 4 (B)	b	g	c	DP
23	68	SEG18	CH3 sign/label	minus	CH3 label	OPEN/ERR	spare
24	67	SEG19	CH3 digit 1 (A)	a	f	e	d
25	66	SEG20	CH3 digit 1 (B)	b	g	c	DP
26	65	SEG21	CH3 digit 2 (A)	a	f	e	d
27	64	SEG22	CH3 digit 2 (B)	b	g	c	DP
28	60	SEG23	CH3 digit 3 (A)	a	f	e	d
29	59	SEG24	CH3 digit 3 (B)	b	g	c	DP
30	58	SEG25	CH3 digit 4 (A)	a	f	e	d
31	57	SEG26	CH3 digit 4 (B)	b	g	c	DP
32	55	SEG28	CH4 sign/label	minus	CH4 label	OPEN/ERR	spare
33	54	SEG29	CH4 digit 1 (A)	a	f	e	d
34	53	SEG30	CH4 digit 1 (B)	b	g	c	DP
35	52	SEG31	CH4 digit 2 (A)	a	f	e	d
36	46	SEG34	CH4 digit 2 (B)	b	g	c	DP
37	45	SEG35	CH4 digit 3 (A)	a	f	e	d
38	44	SEG36	CH4 digit 3 (B)	b	g	c	DP
39	43	SEG37	CH4 digit 4 (A)	a	f	e	d
40	42	SEG38	CH4 digit 4 (B)	b	g	c	DP
41	41	SEG39	global 1	°C	°F	HOLD	MAX
42	40	SEG40	global 2	MIN	LOG/REC	USB/PC	BATT frame
43	39	SEG41	global 3	BATT bar 1	BATT bar 2	BATT bar 3	spare

COM pins on J7:

J7.1 = COM0 (MCU pin 94)
J7.2 = COM1 (MCU pin 93)
J7.3 = COM2 (MCU pin 92)
J7.4 = COM3 (MCU pin 91)

Digit (2 SEG lines × 4 COM):
line A: COM0=a COM1=f COM2=e COM3=d
line B: COM0=b COM1=g COM2=c COM3=DP

Reserved (not on LCD):

SEG27 → BOOT (P31)
SEG32 → RXD1 (UART1)
SEG33 → TXD1 (UART1)
SEG42 → SDA (I2C)
SEG43 → SCL (I2C)